Notes for March 31

Fields:

A field is a set of numbers with two (binary) operations (usually called addition [+] and multiplication [×]) such that the following properties hold:

Addition:

Property Name	Property Description	Examples
Commutativity	For all numbers a and b $a+b = b+a$	$2+3 = 3+2$
Associativity	For all numbers a, b and c $a+(b+c) = (a+b)+c$	$3+(6+17) = (3+6)+17$
Additive Identity	There exists a number (which will be denoted 0) such that for every number a $a+0 = a$	$7+0 = 7$
Additive Inverses	For every number a there exists an additive inverse (which will be denoted -a) such that $a + -a = 0$	$8 + -8 = 0$

Multiplication:

Property Name	Property Description	Examples
Commutativity	For all numbers a and b $a\times b = b\times a$	$2\times 3 = 3\times 2$
Associativity	For all numbers a, b and c $a\times(b\times c) = (a\times b)\times c$	$3\times(6\times 17) = (3\times 6)\times 17$
Multiplicative Identity	There exists a number (which will be denoted 1) such that for every number a $a\times 1 = a$	$7\times 1 = 7$
Multiplicative Inverses for Non-Zero Numbers	For every non-zero number a there exists an multiplicative inverse (which will be denoted a^{-1}) such that $a \times a^{-1} = 1$	$8 \times 8^{-1} = 1$

Addition and Multiplication:

Property Name	Property Description	Examples
Distributivity	For all numbers a, b and c $a\times(b+c) = a\times b + a\times c$	$2\times(3+4) = 2\times 3 + 2\times 4$

Examples and Non-Examples

A. Whole Numbers (1, 2, 3, 4, . . .) with usual +, × is **NOT** a field

Additive Property	Satisfied	Multiplicative Property	Satisfied
Commutative	T	Commutative	T
Associate	T	Associate	T
Identity	V	Identity	T
Inverse	V	Inverse	V
Addition/Multiplication		Satisfied	
Distributive Property		T	

1. There is not an additive identity element (0) in the set of whole numbers.
2. Because there are no negatives in the set of whole numbers there are not any additive inverses in the set of whole numbers – for example 5 does not have an additive inverse in the set of whole numbers
3. Because there are no fractions in the set of whole numbers there are not, in general, any multiplicative inverses in the set of whole numbers – for example 5 does not have an multiplicative inverse in the set of whole numbers

B. Integers (. . . -4, -3, -2, 0, 1, 2, 3, 4, . . .) with usual +, × is **NOT** a field

Additive Property	Satisfied	Multiplicative Property	Satisfied
Commutative	T	Commutative	T
Associate	T	Associate	T
Identity	T	Identity	T
Inverse	T	Inverse	V
Addition/Multiplication		Satisfied	
Distributive Property		T	

1. Because there are no fractions in the set of integers there are not, in general, any multiplicative inverses in the set of integers – for example 5 does not have an multiplicative inverse in the set of integers

C. Real Numbers with usual +, × **IS** a field

Additive Property	Satisfied	Multiplicative Property	Satisfied
Commutative	T	Commutative	T
Associate	T	Associate	T
Identity	T	Identity	T
Inverse	T	Inverse	T
Addition/Multiplication		Satisfied	
Distributive Property		T	

Additionally, the set of rational numbers Q = { p/q | p, q are integers with q ... 0 } with the usual +, × IS a field

D. Plane ú^2 = { (x,y) | x,y are real numbers } with addition and multiplication as follows:

$$(a,b) + (c,d) = (a+c, b+d)$$

$$(a,b) \times (c,d) = (a\times c, b\times d)$$

is **NOT** a field

Additive Property	Satisfied	Multiplicative Property	Satisfied
Commutative	T	Commutative	T
Associate	T	Associate	T
Identity	T (0,0)	Identity	T (1,1)
Inverse	T	Inverse	V
Addition/Multiplication		Satisfied	
Distributive Property		T	

For this multiplication (5,0) is not the "zero" element of the set and it does not have a multiplicative inverse.

E. Complex Plane (Complex Numbers) ÷ = { (x,y) | x,y are real numbers } with addition and multiplication as follows:

Write (a,b) = (a,0) + (0,b) / a + b *i*.

Then, we can think of the horizontal axis which is the set of all { (a,0) | a is real } as being the same as the set of real numbers and we will call it the real axis. Also, we will call the vertical axis which is the set of all { (0,b) = b *i* | b is real } the "imaginary axis". The symbol $i = (0,1)$. We will define multiplication of *i* with itself by $i^2 = -1$. Then:

$$(a,b) + (c,d) = (a+b\,i) + (c+d\,i) = (a+c) + (b+d)\,i = (a+c,\ b+d)$$

$$\begin{aligned}(a,b) \times (c,d) = (a+b\,i) \times (c+d\,i) &= (a\times c + b\times d\,i^2) + (a\times d + b\times c)\,i \\ &= (a\times c - b\times d) + (a\times d + b\times c)\,i \\ &= (a\times c - b\times d,\ a\times d + b\times c)\end{aligned}$$

IS a field.

Additive Property	Satisfied	Multiplicative Property	Satisfied
Commutative	T	Commutative	T
Associate	T	Associate	T
Identity	T	Identity	T
Inverse	T	Inverse	T
Addition/Multiplication		Satisfied	
Distributive Property		T	

Fields are important sets because in a field (real numbers, rational numbers or the complex numbers) all of the usual properties and rules of algebra for manipulating expressions and solving equations are true. For example, in a field there is a property that

$a \times b = 0$ implies that either $a = 0$ or else $b = 0$ (or both)

Another example,

a×b = a×c and a ...0 implies b = c (cancellation rule)

History of Number Systems and Solving Equations:

Whole numbers:

History: Whole numbers were the first numbers used because they represented the process of counting (sheep, wives, children, rocks, etc.) They were associated with the notion of the actual physical amount or presence of a given quantity.

Equations: Linear equations of the form $x = a$ could be solved using only whole numbers, but not equations of the form $x + a = 0$ in this number system. (Here a is a whole number.)

Integers:

History: As society became more complex and interactions between individuals developed, so did the need for expanding the types of numbers needed to represent information. To represent, say debt, negative numbers were needed. The morally weighted names "positive" and "negative" were imposed on numbers to say something about the perceived reality of them.

Equations: Linear equations of the form $x + a = 0$ could be solved using only integers, but not equations of the form $a x + b = 0$ in this number system. (Here a, b are integers).

Rational Numbers:

History: As society became more complex, its needs to represent fractions or proportions as numbers developed, e.g., a father's estate being divided among his heirs; surveying to establish property lines in the Nile valley after the annual spring floods.

Equations: Linear equations of the form $a x + b = 0$ could be solved using only rational numbers, but not quadratic equations like $x^2 = 2$ in this number system. (Here a, b are integers.)

Real Numbers:

History: As the Greeks developed geometry, they courted a philosophy that rational numbers (to which they assigned moral values) could be used to describe the universe. However, with the development of the Pythagorean Theorem, they came to a crisis. They could show that certain physical, measurable lengths could **not** be rational. (In a 45°-45°-90° triangle with two sides of length 1, the hypotenuse must be length $\sqrt{2}$.) The morally weighted names "rational" and "irrational" were imposed on numbers to say something about the perceived reality of them. At this point, the present day notion of the real number line existed and was valid: any length measurement along a scaled, ordered line corresponds to a real number and any real number corresponds to a length measurement along a scaled, ordered line.

Equations: Quadratic equations like $x^2 = 2$ could be solved, but not equations like $x^2 + 1 = 0$ in this number system.

Complex Numbers:

History: To represent solutions for complex problems arising in physics, mechanics and astronomy, the need arose to extend the type of available numbers or else certain equations which represented real world problems would not be solvable. The morally weighted names "real" and "imaginary" were imposed on numbers to say something about the perceived reality of them.

Equations: Quadratic equations like $x^2 + 1 = 0$ could be solved. Given the above progression of needing to add new (abstract) sets of numbers to our existing number system every time a new type of equation was introduced could suggest that if we next wanted to solve other types of quadratics or cubics or quartics (and so forth) that we would again each time need to expand our number system. However, the Fundamental Theorem of Algebra was proved (discovered) which in effect says that we can stop, that at this point we have a complete enough set of numbers in which solutions for every type of equation can be found.

Fundamental Theorem of Algebra:

Every non-constant polynomial $p(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_2 x^2 + a_1 x + a_0$ has a root (in ÷). (Here the coefficients a_n, a_{n-1}, . . . , a_2, a_1, a_0 are integers or real numbers or complex numbers.)

Alternate version.

Every non-constant polynomial $p(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_2 x^2 + a_1 x + a_0$ has a linear factor of the form (a x + b), with a … 0, in its factorization. (Here again the coefficients a_n, a_{n-1}, . . . , a_2, a_1, a_0 are integers or real numbers or complex numbers.)

A root or a zero or a solution of a polynomial $p(x)$ – all three terms mean the same thing – is a value x that solves the equations $p(x) = 0$, that is, is a number x which when substituted in makes $p(x) = 0$.

Examples

Let $p(x) = x^2 - 4x - 5$. Then, the number 5 is a root of $p(x)$, because $p(5) = 0$.

Let $p(x) = x^3 + 1$. Then, the number -1 is a root of $p(x)$, because $p(-1) = 0$.

Let $p(x) = x^2 - 2x + 5$. Then, the number $1 + 2i$ is a root of $p(x)$, because $p(1 + 2i) = 0$.

A linear factor is a factor for the form (a x + b). Note that when attempting to solve a linear equation of the form a x + b = 0, with a … 0, it is always possible to find a solution, namely, x = -b/a.

What the Fundamental Theorem of Algebra does **not** say, is that the root of $p(x)$ will be in the same number system as the coefficients of $p(x)$. Usually, that is not true.

Examples

Let $p(x) = x^2 - 6$. Then, while the coefficients of $p(x)$ are integers, the roots are irrational numbers ($\sqrt{6}$, -$\sqrt{6}$).

Let $p(x) = x^2 + 1$. Then, again while the coefficients of $p(x)$ are integers, the roots are complex numbers (i, $-i$).

One of the first consequences of the Fundamental Theorem of Algebra is that every non-constant polynomial $p(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_2 x^2 + a_1 x + a_0$ of degree n (the degree of a polynomial is the degree of its highest exponent), has exactly n roots. *Alternate version*. Every non-constant polynomial $p(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_2 x^2 + a_1 x + a_0$ of degree n (the degree of a polynomial is the degree of its highest exponent), can be factored into a product of exactly n linear factors.

A quadratic polynomial has always exactly 2 roots. A cubic polynomial has always exactly 3 roots. A fifth degree polynomial has always exactly 5 roots. Etc.

Examples

Let $p(x) = x^2 - 4x - 5$. Then, $p(x)$ has 2 roots (5 and -1).

Let $p(x) = x^3 + 1$. Then, $p(x)$ has 3 roots (-1, ½ (1+$\sqrt{3}\,i$), ½ (1-$\sqrt{3}\,i$)).

Let $p(x) = x^2 - 2x + 5$. Then, $p(x)$ has 2 roots (1 + 2 i , 1 - 2 i).

Finally, one last note about the roots of polynomials. Note that if a + b i is a complex number we call a - b i the complex conjugate. From the Theory of Equations we have the following:

Theorem: Let $p(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_2 x^2 + a_1 x + a_0$ be a polynomial with **real** coefficients. Then, any complex roots of $p(x)$ must occur in conjugate pairs.

Examples

Let $p(x) = x^2 - 4x - 5$. The theorem does not apply because in this case $p(x)$ does not happen to have any complex roots.

Let $p(x) = x^3 + 1$. Then, ½ (1+$\sqrt{3}\,i$) is a complex root of $p(x)$ and so is its conjugate ½ (1-$\sqrt{3}\,i$)

Let $p(x) = x^2 - 2x + 5$. Then, 1 + 2 i is a complex root of $p(x)$ and so is its conjugate 1 - 2 i